

ADV: Financial Maths (Adv), M1 Financial Modelling (Adv)
Arithmetic Series (Y12)

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Exam Equivalent Time: 118.5 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

HISTORICAL CONTRIBUTION

- *M1 Financial Modelling* is a Year 12 topic that contains the substantial *Series and Application* topic of the old Mathematics course, and adds common Standard 2 content on loan and annuity tables.
- The topic is split into four sub-topics for the purposes of analysis: *1-Modelling Investments and Loans*, *2-Arithmetic Series*, *3-Geometric Series* and *4-Financial Applications of Series*.
- This analysis looks at *Arithmetic Series*.

HSC ANALYSIS - What to expect and common pitfalls

- *Arithmetic Series* was last examined in the 2020 exam in a question that involved standard calculations of formulae in a non-applied context.
- Questions combining arithmetic series with financial themes should be a revision focus - the database includes applied contexts such as wages and depreciation that cover this area.
- All past HSC questions on this sub-topic have been included in the database whether they involve a financial context or not, as they still represent high quality revision in our view.
- The recursive definition of an arithmetic sequence is covered in response to syllabus changes. We recommend giving attention to *M1 SM-Bank 7* where the straight line depreciation is modelled in this way.
- *Arithmetic Series* is a great topic area to score highly, with the notable exceptions of *2018 Adv 14d* and *2013 Adv 12c* where a combination of arithmetic and geometric series caused significant problems - these deserve attention.

Questions

1. Financial Maths, 2ADV M1 2015 HSC 3 MC

The first three terms of an arithmetic series are 3, 7 and 11.

What is the 15th term of this series?

- (A) 59
(B) 63
(C) 465
(D) 495

2. Financial Maths, 2ADV M1 SM-Bank 1 MC

On day 1, Vikki spends 90 minutes on a training program.

On each following day, she spends 10 minutes less on the training program than she did the day before.

Let t_n be the number of minutes that Vikki spends on the training program on day n .

A recursive equation that can be used to model this situation for $1 \leq n \leq 10$ is

- A. $t_{n+1} = 0.90t_n$ $t_1 = 90$
B. $t_{n+1} = 1.10t_n$ $t_1 = 90$
C. $t_{n+1} = 1 - 10t_n$ $t_1 = 90$
D. $t_{n+1} = t_n - 10$ $t_1 = 90$

3. Financial Maths, 2ADV M1 SM-Bank 8 MC

There are 3000 tickets available for a concert.

On the first day of ticket sales, 200 tickets are sold.

On the second day, 250 tickets are sold.

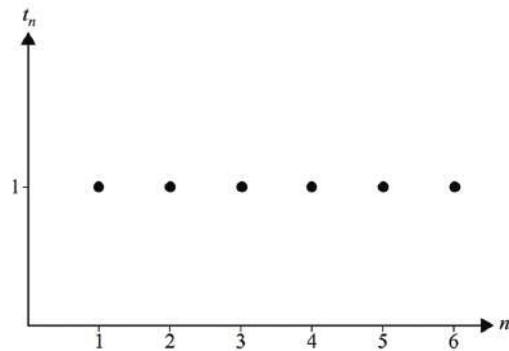
On the third day, 300 tickets are sold.

This pattern of ticket sales continues until all 3000 tickets are sold.

How many days does it take for all of the tickets to be sold?

- (A) 5
(B) 6
(C) 8
(D) 34
-

4. Financial Maths, 2ADV M1 SM-Bank 10 MC



The graph above shows the first six terms of a sequence.

This sequence could be

- (A) an arithmetic sequence that sums to one.
- (B) an arithmetic sequence with a common difference of one.
- (C) a geometric sequence with an infinite sum of one.
- (D) a geometric sequence with a common ratio of one.

5. Financial Maths, 2ADV M1 SM-Bank 5 MC

The first three terms of an arithmetic sequence are **1, 3, 5...**

The sum of the first n terms of this sequence, S_n , is

- (A) $S_n = n^2$
- (B) $S_n = 2n$
- (C) $S_n = 2n - 1$
- (D) $S_n = 2n + 1$

6. Financial Maths, 2ADV M1 2011 HSC 3a

A skyscraper of 110 floors is to be built. The first floor to be built will cost \$3 million. The cost of building each subsequent floor will be \$0.5 million more than the floor immediately below.

- i. What will be the cost of building the 25th floor? **(2 marks)**
- ii. What will be the cost of building all 110 floors of the skyscraper? **(2 marks)**

7. Financial Maths, 2ADV M1 2020 HSC 12

Calculate the sum of the arithmetic series **$4 + 10 + 16 + \dots + 1354$** . **(3 marks)**

8. Financial Maths, 2ADV M1 2021 HSC 14

The first term of an arithmetic sequence is 5. The sum of the first 43 terms is 2021.

What is the common difference of the sequence?

9. Financial Maths, 2ADV M1 SM-Bank 7

Joe buys a tractor under a buy-back scheme. This scheme gives Joe the right to sell the tractor back to the dealer.

The recurrence relation below can be used to calculate the price Joe sells the tractor back to the dealer (P_n), after n years

$$P_0 = 56\,000, \quad P_n = P_{n-1} - 7000$$

- i. Write the general rule to find the value of P_n in terms of n . **(1 mark)**
- ii. After how many years will the dealer offer to buy back Joe's tractor at half of its original value. **(1 mark)**

10. Financial Maths, 2ADV M1 2009 HSC 3a

An arithmetic series has 21 terms. The first term is 3 and the last term is 53.

Find the sum of the series. **(2 marks)**

11. Financial Maths, 2ADV M1 2005 HSC 3a

Evaluate $\sum_{n=3}^5 (2n + 1)$. **(1 mark)**

12. Financial Maths, 2ADV M1 2006 HSC 3c

On the first day of the harvest, an orchard produces 560 kg of fruit. On the next day, the orchard produces 543 kg, and the amount produced continues to decrease by the same amount each day.

- i. How much fruit is produced on the fourteenth day of the harvest? **(2 marks)**
- ii. What is the total amount of fruit that is produced in the first 14 days of the harvest? **(1 mark)**
- iii. On what day does the daily production first fall below 60 kg? **(2 marks)**

13. Financial Maths, 2ADV M1 2018 HSC 11d

In an arithmetic series, the third term is 8 and the twentieth term is 59.

- i. Find the common difference. **(1 mark)**
- ii. Find the 50th term. **(2 marks)**

14. Financial Maths, 2ADV M1 2008 HSC 1f

Find the sum of the first 21 terms of the arithmetic series $3 + 7 + 11 + \dots$ (2 marks)

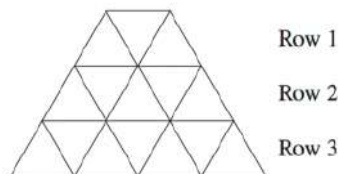
15. Financial Maths, 2ADV M1 2019 HSC 12b

In an arithmetic series, the fourth term is 6 and the sum of the first 16 terms is 120.

Find the common difference. (3 marks)

16. Financial Maths, 2ADV M1 2012 HSC 12c

Jay is making a pattern using triangular tiles. The pattern has 3 tiles in the first row, 5 tiles in the second row, and each successive row has 2 more tiles than the previous row.



- How many tiles would Jay use in row 20? (2 marks)
 - How many tiles would Jay use altogether to make the first 20 rows? (1 mark)
 - Jay has only 200 tiles. How many complete rows of the pattern can Jay make? (2 marks)
-

17. Financial Maths, 2ADV M1 2010 HSC 4a

Susanna is training for a fun run by running every week for 26 weeks. She runs 1 km in the first week and each week after that she runs 750 m more than the previous week, until she reaches 10 km in a week. She then continues to run 10 km each week.

- How far does Susannah run in the 9th week? (1 mark)
 - In which week does she first run 10 km? (1 mark)
 - What is the total distance that Susannah runs in 26 weeks? (2 marks)
-

18. Financial Maths, 2ADV M1 2004 HSC 5a

Clare is learning to drive. Her first lesson is 30 minutes long. Her second lesson is 35 minutes long. Each subsequent lesson is 5 minutes longer than the lesson before.

- How long will Clare's twenty-first lesson be? (1 mark)
 - How many hours of lessons will Clare have completed after her twenty-first lesson? (2 marks)
 - During which lesson will Clare have completed a total of 50 hours of driving lessons? (2 marks)
-

19. Financial Maths, 2ADV M1 2005 HSC 7a

Anne and Kay are employed by an accounting firm.

Anne accepts employment with an initial annual salary of \$50 000. In each of the following years her annual salary is increased by \$2500.

Kay accepts employment with an initial annual salary of \$50 000. In each of the following years her annual salary is increased by 4%.

- What is Anne's annual salary in her thirteenth year? (2 marks)
 - What is Kay's annual salary in her thirteenth year? (2 marks)
 - By what amount does the total amount paid to Kay in her first twenty years exceed that paid to Anne in her first twenty years? (3 marks)
-

20. Financial Maths, 2ADV M1 2007 HSC 3b

Heather decides to swim every day to improve her fitness level.

On the first day she swims 750 metres, and on each day after that she swims **100** metres more than the previous day. That is, she swims 850 metres on the second day, 950 metres on the third day and so on.

- Write down a formula for the distance she swims on the n th day. (1 mark)
 - How far does she swim on the 10th day? (1 mark)
 - What is the total distance she swims in the first 10 days? (1 mark)
 - After how many days does the total distance she has swum equal the width of the English Channel, a distance of 34 kilometres? (2 marks)
-

21. Financial Maths, 2ADV M1 2014 HSC 12a

Evaluate the arithmetic series $2 + 5 + 8 + 11 + \dots + 1094$. (2 marks)

22. Financial Maths, 2ADV M1 2017 HSC 12c

In an arithmetic series, the fifth term is 200 and the sum of the first four terms is 1200.

Find the value of the tenth term. (3 marks)

23. Financial Maths, 2ADV M1 2018 HSC 14d

An artist posted a song online. Each day there were $2^n + n$ downloads, where n is the number of days after the song was posted.

- Find the number of downloads on each of the first 3 days after the song was posted. (1 mark)
 - What is the total number of times the song was downloaded in the first 20 days after it was posted? (2 marks)
-

24. Financial Maths, 2ADV M1 2013 HSC 12c

Kim and Alex start jobs at the beginning of the same year. Kim's annual salary in the first year is \$30,000 and increases by 5% at the beginning of each subsequent year. Alex's annual salary in the first year is \$33,000, and increases by \$1,500 at the beginning of each subsequent year.

- Show that in the 10th year, Kim's annual salary is higher than Alex's annual salary. (2 marks)
- In the first 10 years how much, in total, does Kim earn? (2 marks)
- Every year, Alex saves $\frac{1}{3}$ of her annual salary. How many years does it take her to save \$87,500? (3 marks)

25. Financial Maths, 2ADV M1 SM-Bank 6

A toy train track consists of a number of pieces of track which join together.

The shortest piece of the track is 15 centimetres long and each piece of track after the shortest is 2 centimetres longer than the previous piece.

The total length of the complete track is 7.35 metres.

Find the length of the longest piece of track. (3 marks)

26. Financial Maths, 2ADV M1 SM-Bank 12

There are 10 checkpoints in a 4500 metre orienteering course. Checkpoint 1 is the start and checkpoint 10 is the finish.

The distance between successive checkpoints increases by 50 metres as each checkpoint is passed.

Calculate the distance, in metres, between checkpoint 2 and checkpoint 3. (3 marks)

Worked Solutions

1. Financial Maths, 2ADV M1 2015 HSC 3 MC

$$3, 7, 11, \dots$$

$$T_1 = 3$$

$$T_2 = 7$$

$$\Rightarrow \text{AP where } a = 3, d = 7 - 3 = 4$$

$$T_n = a + (n - 1)d$$

$$T_{15} = 3 + (15 - 1)4$$

$$= 59$$

$$\Rightarrow A$$

2. Financial Maths, 2ADV M1 SM-Bank 1 MC

Difference equation where each term is 10 minutes less than the preceding term.

$$\therefore \text{Equation } t_{n+1} = t_n - 10, t_1 = 90$$

$$\Rightarrow D$$

3. Financial Maths, 2ADV M1 SM-Bank 8 MC

$$200 + 250 + 300 + \dots$$

AP where $a = 200$

$$d = 250 - 200 = 50$$

Find n when $S_n = 3000$:

$$S_n = \frac{n}{2}[2a + (n-1)d]$$

$$3000 = \frac{n}{2}[2 \times 200 + (n-1)50]$$

$$= \frac{n}{2}(400 + 50n - 50)$$

$$= \frac{n}{2}(350 + 50n)$$

$$3000 = 175n + 25n^2$$

$$n^2 + 7n - 120 = 0$$

$$(n-8)(n+15) = 0$$

$$\therefore n = 8, \quad (n > 0)$$

$$\Rightarrow C$$

4. Financial Maths, 2ADV M1 SM-Bank 10 MC

Series is 1, 1, 1, ...

The only possibility within the choices

is a geometric sequence where $r = 1$.

$$\Rightarrow D$$

5. Financial Maths, 2ADV M1 SM-Bank 5 MC

Sequence is 1, 3, 5, ...

AP where $a = 1$, and

$$d = 3 - 1 = 2$$

$$S_n = \frac{n}{2}[2a + (n-1)d]$$

$$= \frac{n}{2}[2 \times 1 + (n-1)2]$$

$$= \frac{n}{2}[2 + 2n - 2]$$

$$= n^2$$

$$\Rightarrow A$$

6. Financial Maths, 2ADV M1 2011 HSC 3a

i. $T_1 = a = 3$

$$T_2 = a + d = 3.5$$

$$T_3 = a + 2d = 4$$

$$\Rightarrow \text{AP where } a = 3 \quad d = 0.5$$

$$T_{25} = a + 24d$$

$$= 3 + 24(0.5)$$

$$= 15$$

$$\therefore \text{The 25th floor costs } \$15,000,000.$$

MARKER'S COMMENT: Better responses listed the sequence of terms in the series as illustrated.

ii. $S_{110} = \text{Total cost of 110 floors}$

$$= \frac{n}{2}(2a + (n-1)d)$$

$$= \frac{110}{2}(2 \times 3\,000\,000 + (110-1)500\,000)$$

$$= 55(6\,000\,000 + 49\,500\,000)$$

$$= \$3327.5 \text{ million}$$

$$\therefore \text{The total cost of 110 floors is } \$3327.5 \text{ million}$$

MARKER'S COMMENT: Better responses stated the formula *BEFORE* any calculations were performed. This allows markers to allocate part marks to students who had errors in calculation.

7. Financial Maths, 2ADV M1 2020 HSC 12

$$a = 4, l = 1354, d = 10 - 4 = 6$$

Find n :

$$T_n = a + (n - 1)d$$

$$1354 = 4 + (n - 1)6$$

$$1354 = 6n - 2$$

$$n = \frac{1356}{6}$$

$$= 226$$

$$\begin{aligned}\therefore S_{226} &= \frac{n}{2}(a + l) \\ &= \frac{226}{2}(4 + 1354) \\ &= 153\,454\end{aligned}$$

8. Financial Maths, 2ADV M1 2021 HSC 14

$$T_1 = a = 5$$

$$S_{43} = \frac{n}{2}[2a + (n - 1)d]$$

$$2021 = \frac{43}{2}(10 + 42d)$$

$$2021 = 215 + 903d$$

$$903d = 1806$$

$$\therefore d = 2$$

9. Financial Maths, 2ADV M1 SM-Bank 7

i. $P_1 = P_0 - 7000$

$$\begin{aligned}P_2 &= P_0 - 7000 - 7000 \\ &= 56\,000 - 7000 \times 2\end{aligned}$$

$$\vdots$$

$$P_n = 56\,000 - 7000n$$

ii. Half original value = $56\,000 \div 2 = \$28\,000$

Find n such that:

$$28\,000 = 56\,000 - 7000n$$

$$7000n = 28\,000$$

$$\therefore n = 4 \text{ years}$$

10. Financial Maths, 2ADV M1 2009 HSC 3a

$$S_n = \frac{n}{2}(a + l)$$

$$\begin{aligned}S_{21} &= \frac{21}{2}(3 + 53) \\ &= 588\end{aligned}$$

11. Financial Maths, 2ADV M1 2005 HSC 3a

$$\begin{aligned}\sum_{n=3}^5 (2n + 1) &= (2 \times 3 + 1) + (2 \times 4 + 1) + (2 \times 5 + 1) \\ &= 7 + 9 + 11 \\ &= 27\end{aligned}$$

12. Financial Maths, 2ADV M1 2006 HSC 3c

i. $T_1 = a = 560$

$$T_2 = a + d = 543$$

$$\Rightarrow \text{AP where } a = 560, d = -17$$

$\vdots$

$$T_{14} = a + 13d$$

$$= 560 - (13 \times 17)$$

$$= 339$$

$\therefore$ 339 kg of fruit is produced on 14th day.

ii. $S_{14} = \text{total fruit produced in 1st 14 days}$

$$S_{14} = \frac{n}{2}[2a + (n-1)d]$$

$$= \frac{14}{2}[2 \times 560 - (14-1) \times 17]$$

$$= 7[1120 - 221]$$

$$= 6293$$

$\therefore$ 6293 kg of fruit is produced in the 1st 14th days.

iii. Find n such that $T_n < 60$

$$T_n = a + (n-1)d < 60$$

$$560 - 17(n-1) < 60$$

$$560 - 17n + 17 < 60$$

$$17n > 517$$

$$n > 30.41\dots$$

$\therefore$ On the 31st day, production will first drop below 60 kg.

13. Financial Maths, 2ADV M1 2018 HSC 11d

i. $a + 2d = 8 \quad \dots (1)$

$$a + 19d = 59 \quad \dots (2)$$

Subtract $(2) - (1)$

$$17d = 51$$

$$\therefore d = 3$$

ii. Find T_{50}

Substitute $d = 3$ into (1)

$$\Rightarrow a = 12$$

$$T_n = a + (n-1)d$$

$$\therefore T_{50} = 2 + 49 \times 3$$

$$= 149$$

14. Financial Maths, 2ADV M1 2008 HSC 1f

$$S = 3 + 7 + 11 + \dots$$

$$a = 3$$

$$d = 7 - 3 = 4$$

$$\therefore S_{21} = \frac{n}{2}[2a + (n-1)d]$$

$$= \frac{21}{2}[2 \times 3 + (21-1)4]$$

$$= \frac{21}{2}[6 + 80]$$

$$= 903$$

15. Financial Maths, 2ADV M1 2019 HSC 12b

$$T_4 = 6,$$

$$a + 3d = 6 \dots (1)$$

$$S_{16} = 120,$$

$$\frac{16}{2}(2a + 15d) = 120$$

$$16a + 120d = 120 \dots (2)$$

Substitute $a = 6 - 3d$ from (1) into (2):

$$16(6 - 3d) + 120d = 120$$

$$96 - 48d + 120d = 120$$

$$72d = 24$$

$$d = \frac{1}{3}$$

16. Financial Maths, 2ADV M1 2012 HSC 12c

i. $T_1 = a = 3$

$$T_2 = a + d = 5$$

$$T_3 = a + 2d = 7$$

$$\Rightarrow \text{AP where } a = 3, d = 2$$

$\vdots$

$$T_{20} = a + 19d$$

$$= 3 + 19(2)$$

$$= 41$$

$\therefore$ Row 20 has 41 tiles.

ii. $S_{20} =$ the total number of tiles in first 20 rows

$$S_{20} = \frac{n}{2}(a + l)$$

$$= \frac{20}{2}(3 + 41)$$

$$= 440$$

$\therefore$ There are 440 tiles in the first 20 rows.

iii. If Jay only has 200 tiles, then $S_n \leq 200$

$$\frac{n}{2}(2a + (n - 1)d) \leq 200$$

$$\frac{n}{2}(6 + 2n - 2) \leq 200$$

$$n(n + 2) \leq 200$$

$$n^2 + 2n - 200 \leq 0$$

$$n = \frac{-2 \pm \sqrt{4 + 4 \cdot 1 \cdot 200}}{2 \cdot 1}$$

$$= \frac{-2 \pm \sqrt{804}}{2}$$

$$= -1 \pm \sqrt{201}$$

$$= 13.16 \text{ (answer must be positive)}$$

MARKER'S COMMENT: Better responses stated the formula BEFORE any calculations were performed. This enabled students to get some marks if they made an error in their working.

NOTE: Examiners often ask questions requiring n to be found using the formula

$$S_n = \frac{n}{2}[2a + (n - 1)d]$$

as this requires the solving of a quadratic, and interpretation of the answer.

$\therefore$ Jay can complete 13 rows.

17. Financial Maths, 2ADV M1 2010 HSC 4a

i. $T_1 = a = 1$

$$T_2 = a + d = 1.75$$

$$T_3 = a + 2d = 2.50$$

$$\Rightarrow \text{AP where } a = 1 \quad d = 0.75$$

$\vdots$

$$T_9 = a + 8d$$

$$= 1 + 8(0.75)$$

$$= 7$$

$\therefore$ Susannah runs 7 km in the 9th week.

ii. Find n such that $T_n = 10$ km

$$\text{Using } T_n = a + (n - 1)d$$

$$1 + (n - 1)(0.75) = 10$$

$$0.75n - 0.75 = 9$$

$$n = \frac{9.75}{0.75}$$

$$= 13$$

MARKER'S COMMENT: Better responses wrote the formula for the n th term before clearly substituting in known values a and d .

$\therefore$ Susannah runs 10 km for the first time in the 13th Week.

iii. Let D = the total distance Susannah runs in 26 weeks

$$D = S_{13} + 13(10)$$

$$= \frac{n}{2}[2a + (n - 1)d] + 13(10)$$

$$= \frac{13}{2}[2(1) + (13 - 1)(0.75)] + 130$$

$$= \frac{13}{2}(2 + 9) + 130$$

$$= 201.5$$

$\therefore$ Susannah runs a total of 201.5 km in 26 weeks.

MARKER'S COMMENT: Many students incorrectly calculated S_{26} , not taking into account the AP stopped at the 13th term.

18. Financial Maths, 2ADV M1 2004 HSC 5a

i. 30, 35, 40, ...

$\Rightarrow$ AP where $a = 30$, $d = 5$

$$T_n = a + (n - 1)d$$

$$T_{21} = 30 + 20(5)$$

$$= 30 + 100$$

$$= 130$$

$\therefore$ Clare's twenty-first lesson will be 130 minutes long.

ii. Find S_{21} given $a = 30$, $d = 5$

$$S_n = \frac{n}{2}[2a + (n - 1)d]$$

$$\therefore S_{21} = \frac{21}{2}[2(30) + 20 \times 5]$$

$$= \frac{21}{2}[60 + 100]$$

$$= \frac{21}{2} \times 160$$

$$= 1680 \text{ minutes}$$

$$= 28 \text{ hours.}$$

(iii) 50 hours = 50×60

$$= 3000 \text{ minutes}$$

Find n , given $a = 30$, $d = 5$, $S_n = 3000$

$$S_n = \frac{n}{2}[2a + (n - 1)d]$$

$$\therefore 3000 = \frac{n}{2}[2(30) + (n - 1)5]$$

$$6000 = n[60 + 5n - 5]$$

$$6000 = n[55 + 5n]$$

$$6000 = 55n + 5n^2$$

$$5n^2 + 55n - 6000 = 0$$

$$n^2 + 11n - 1200 = 0$$

$$\begin{aligned}\therefore n &= \frac{-11 \pm \sqrt{(11)^2 - 4(1)(-1200)}}{2(1)} \\ &= \frac{-11 \pm \sqrt{121 + 4800}}{2} \\ &= \frac{-11 \pm \sqrt{4921}}{2} \\ &= 29.5749... \quad (n > 0)\end{aligned}$$

$\therefore$ Clare completes 50 hours of lessons during her 30th lesson.

19. Financial Maths, 2ADV M1 2005 HSC 7a

i. Let T_n = Anne's salary in year n

$$T_1 = a = \$50\,000$$

$$T_2 = a + d = \$52\,500$$

$$\Rightarrow \text{AP where } a = \$50\,000, \quad d = \$2500$$

$$T_n = a + (n - 1)d$$

$$T_{13} = 50\,000 + (13 - 1) \times 2500$$

$$= 80\,000$$

$\therefore$ Anne's salary in her 13th year is \$80 000.

ii. Let K_1 = Kay's salary in year n

$$K_1 = a = 50\,000$$

$$K_2 = ar = 50\,000 \times 1.04 = 52\,000$$

$$\Rightarrow \text{GP where } a = 50\,000, \quad r = 1.04$$

$$K_n = ar^{n-1}$$

$$K_{13} = 50\,000 \times (1.04)^{12}$$

$$= \$80\,051.61\dots$$

$$= \$80\,052 \quad (\text{nearest dollar})$$

iii. Anne

$$S_n = \frac{n}{2}[2a + (n - 1)d]$$

$$S_{20} = \frac{20}{2}[2 \times 50\,000 + (20 - 1)2500]$$

$$= 10[100\,000 + 47\,500]$$

$$= \$1\,475\,000$$

Kay

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$S_{20} = \frac{50\,000(1.04^{20} - 1)}{1.04 - 1}$$

$$= \$1\,488\,903.929\dots$$

Difference

$$= 1\,488\,903.929\dots - 1\,475\,000$$

$$= \$13\,903.928\dots$$

$$= \$13\,904 \quad (\text{nearest \$})$$

$\therefore$ Kay's total salary exceeds Anne's by \$13 904

20. Financial Maths, 2ADV M1 2007 HSC 3b

i. $T_1 = 750$

$T_2 = 850$

$\Rightarrow$ AP where $a = 750$, $d = 100$

$$\begin{aligned}\therefore T_n &= a + (n - 1)d \\ &= 750 + (n - 1)100 \\ &= 750 + 100n - 100 \\ &= 650 + 100n\end{aligned}$$

ii. $T_{10} = 650 + 100 \times 10$
 $= 1650$ metres

$\therefore$ She swims 1650 metres on the 10th day.

iii. $S_n = \frac{n}{2}[2a + (n - 1)d]$

$$\begin{aligned}\therefore S_{10} &= \frac{10}{2}[2 \times 750 + (10 - 1)100] \\ &= 5[1500 + 900] \\ &= 12\,000\end{aligned}$$

$\therefore$ She swims 12 km in the first 10 days.

iv. Find n such that $S_n = 34$ km

$$\frac{n}{2}[2 \times 750 + (n - 1)100] = 34\,000$$

$$\frac{n}{2}[1500 + 100n - 100] = 34\,000$$

$$\frac{n}{2}[1400 + 100n] = 34\,000$$

$$700n + 50n^2 = 34\,000$$

$$50n^2 + 700n - 34\,000 = 0$$

$$50(n^2 + 14n - 680) = 0$$

Using the quadratic formula

$$n$$

$$\begin{aligned}&= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-14 \pm \sqrt{14^2 - 4 \times 1 \times (-680)}}{2 \times 1} \\ &= \frac{-14 \pm \sqrt{2916}}{2} \\ &= \frac{-14 \pm 54}{2} \\ &= 20 \text{ or } -34 \\ &= 20, n > 0\end{aligned}$$

$\therefore$ Her total distance equals 34 km after 20 days.

21. Financial Maths, 2ADV M1 2014 HSC 12a

$2 + 5 + 8 + \dots + 1094$

AP where $a = 2$, $d = 5 - 2 = 3$

Find n :

$$T_n = a + (n - 1)d$$

$$1094 = 2 + (n - 1)3$$

$$3n - 3 = 1092$$

$$3n = 1095$$

$$n = 365$$

$$\therefore S_{365} = \frac{n}{2}(a + l)$$

$$= \frac{365}{2}(2 + 1094)$$

$$= 200\,020$$

22. Financial Maths, 2ADV M1 2017 HSC 12c

$$T_n = a + (n - 1)d$$

$$\Rightarrow a + 4d = 200 \dots (1)$$

$$S_n = \frac{n}{2}[2a + (n - 1)d]$$

$$\Rightarrow 4a + 6d = 1200 \dots (2)$$

Multiply $(1) \times 4$

$$\Rightarrow 4a + 16d = 800 \dots (1')$$

Subtract $(1') - (2)$

$$10d = -400$$

$$d = -40$$

Substitute $d = -40$ into (1)

$$a - 160 = 200$$

$$a = 360$$

$$\therefore T_{10} = 360 - 9(-40)$$

$$= 0$$

23. Financial Maths, 2ADV M1 2018 HSC 14d

i. **Day 1:** $2^1 + 1 = 3$

Day 2: $2^2 + 2 = 6$

Day 3: $2^3 + 3 = 11$

ii. **Total downloads (20 days)**

$$= 2^1 + 1 + 2^2 + 2 + \dots + 2^{20} + 20$$

$$= \underbrace{2^1 + 2^2 + \dots + 2^{20}}_{\text{GP, } a=2, r=2} + \underbrace{1 + 2 + \dots + 20}_{\text{AP, } a=1, d=1}$$

$$= \frac{2(2^{20} - 1)}{2 - 1} + \frac{20}{2}(1 + 20)$$

$$= 2\,097\,150 + 210$$

$$= 2\,097\,360$$

♦ Mean mark 38%.

24. Financial Maths, 2ADV M1 2013 HSC 12c

i. Let K_n = Kim's salary in Year n

$$\left. \begin{array}{l} K_1 = a = 30\,000 \\ K_2 = ar^1 = 30\,000(1.05^1) \end{array} \right\} \Rightarrow GP$$

$$\left. \begin{array}{l} a = 30\,000 \\ r = 1.05 \end{array} \right\}$$

$\vdots$

$$\therefore K_{10} = ar^9 = 30\,000(1.05)^9 = \$46\,539.85$$

Let A_n = Alex's salary in Year n

$$\left. \begin{array}{l} A_1 = a = 33\,000 \\ A_2 = 33\,000 + 1500 = 34\,500 \end{array} \right\} \Rightarrow AP$$

$$\left. \begin{array}{l} a = 33\,000 \\ d = 1500 \end{array} \right\}$$

$\vdots$

$$A_{10} = a + 9d = 33\,000 + 1500(9) = \$46\,500$$

$$\Rightarrow K_{10} > A_{10}$$

$\therefore$ Kim earns more than Alex in the 10th year

ii. In the first 10 years, Kim earns

$$K_1 + K_2 + \dots + K_{10}$$

$$\begin{aligned} S_{10} &= a \left(\frac{r^n - 1}{r - 1} \right) \\ &= 30\,000 \left(\frac{1.05^{10} - 1}{1.05 - 1} \right) \\ &= 377\,336.78 \end{aligned}$$

$\therefore$ In the first 10 years, Kim earns \$377 336.78

iii. Let T_n = Alex's savings in Year n

$$\left. \begin{array}{l} T_1 = a = \frac{1}{3}(33\,000) = 11\,000 \\ T_2 = a + d = \frac{1}{3}(34\,500) = 11\,500 \\ T_3 = a + 2d = \frac{1}{3}(36\,000) = 12\,000 \end{array} \right\} \Rightarrow AP$$

$$\left. \begin{array}{l} a = 11\,000 \\ d = 500 \end{array} \right\}$$

Find n such that $S_n = 87\,500$

$$S_n = \frac{n}{2}[2a + (n - 1)d]$$

$$87\,500 = \frac{n}{2}[22\,000 + (n - 1)500]$$

$$87\,500 = \frac{n}{2}[21\,500 + 500n]$$

$$250n^2 + 10\,750n - 87\,500 = 0$$

$$n^2 + 43n - 350 = 0$$

$$(n - 7)(n + 50) = 0$$

$$\therefore n = 7, \quad n > 0$$

$\therefore$ Alex's savings will be \$87,500 after 7 years.

25. Financial Maths, 2ADV M1 SM-Bank 6

Sequence is 15, 17, 19, ...

AP where $a = 15$, and

$$d = 2$$

$$S_n = \frac{n}{2}[2a + (n - 1)d]$$

$$\therefore 735 = \frac{n}{2}[2 \times 15 + (n - 1)2]$$

$$= \frac{n}{2}[30 + 2n - 2]$$

$$= n^2 + 14n$$

$$0 = n^2 + 14n - 735$$

Using the quadratic formula

$$n = \frac{-14 \pm \sqrt{14^2 - 4 \cdot 1 \cdot (-735)}}{2 \times 1}$$

$$= \frac{-14 \pm 56}{2}$$

$$= 21 \quad (n > 0)$$

$$\begin{aligned} \therefore \text{Longest piece of track} &= a + (n - 1)d \\ &= 15 + (21 - 1)2 \\ &= 55 \text{ cm} \end{aligned}$$

♦ Mean mark 45%.

IMPORTANT: Using the AP sum formula to create and then solve a quadratic in n is challenging and often examined. Students need to solve and interpret the solutions.

26. Financial Maths, 2ADV M1 SM-Bank 12

9 intervals exist between 10 checkpoints

$\Rightarrow$ 9 distances form an AP

Where $d = 50$ $S_9 = 4500$

$$S_n = \frac{n}{2}[2a + (n - 1)d]$$

$$4500 = \frac{9}{2}[2a + (9 - 1) \times 50]$$

$$4500 = \frac{9}{2}[2a + 400]$$

$$9a = 2700$$

$$a = 300$$

Distance between checkpoint 1 and 2

$$= a = 300 \text{ m}$$

$\therefore$ Distance between checkpoint 2 and 3

$$= a + d = 350 \text{ m}$$